

# Mount Kenya University

## Unlocking Infinite Possibilities

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**ADM NO:** BSCCS/2024/54025

**COURSE:** BACHELOR OF SCIENCE IN COMPUTER SCIENCE.

**UNIT TITLE:** BIT 3208.

**UNIT NAME :**Advanced Web Design And Development

**LECTURER :** MR. MICHAEL NYORO.

**SEMESTER:**(Year III, Sem II)

### PROJECT DETAILS

Project title: **PIVOT HOME**

#### Selected Technologies

**Backend:** PHP,MySQL,MySQLi,Apache

**Frontend:** HTML,CSS,Javascript

### **Online Portfolio / GitHub Link**

<https://github.com/5star360/furniture-store>

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# Week 1: Local Environment Setup

## TITLE

## Installation and Testing of Local Development Environment

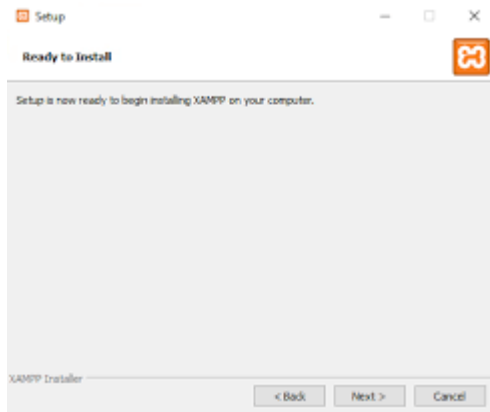


Fig 1: XAMPP Installation.

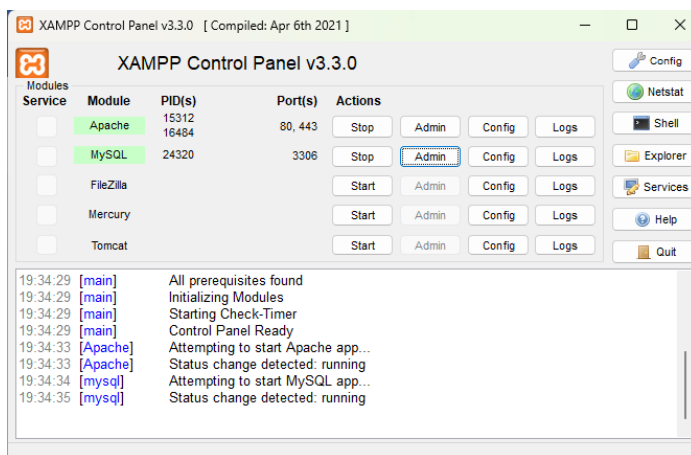


Fig 2: Apache and MySQL Running.

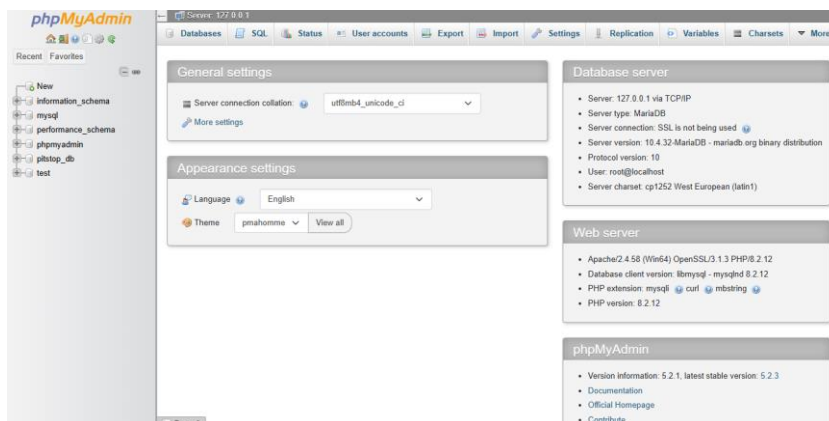


Fig 3: XAMPP Dashboard Page.

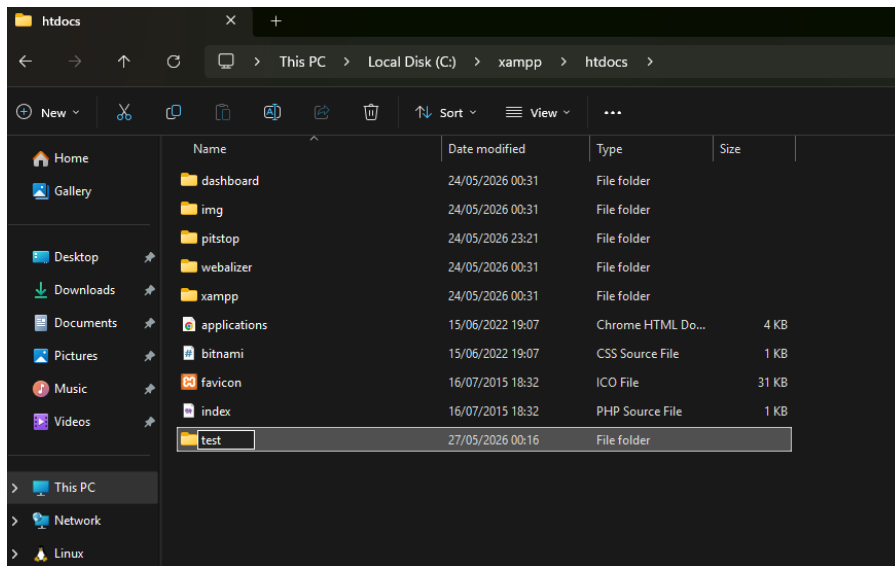


Fig 4: Test folder inside XAMPP htdocs folder.

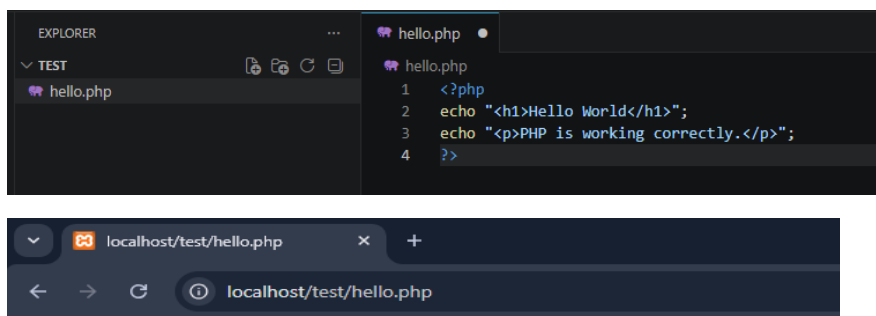
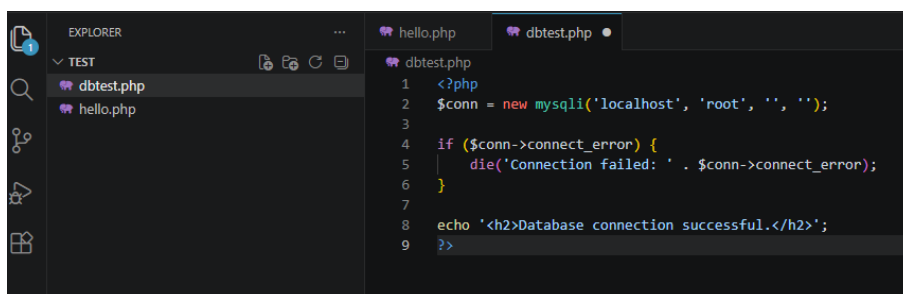


Fig 5: First PHP Page.



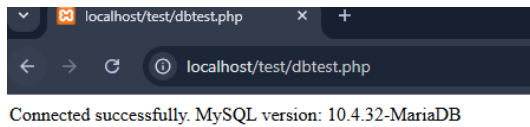


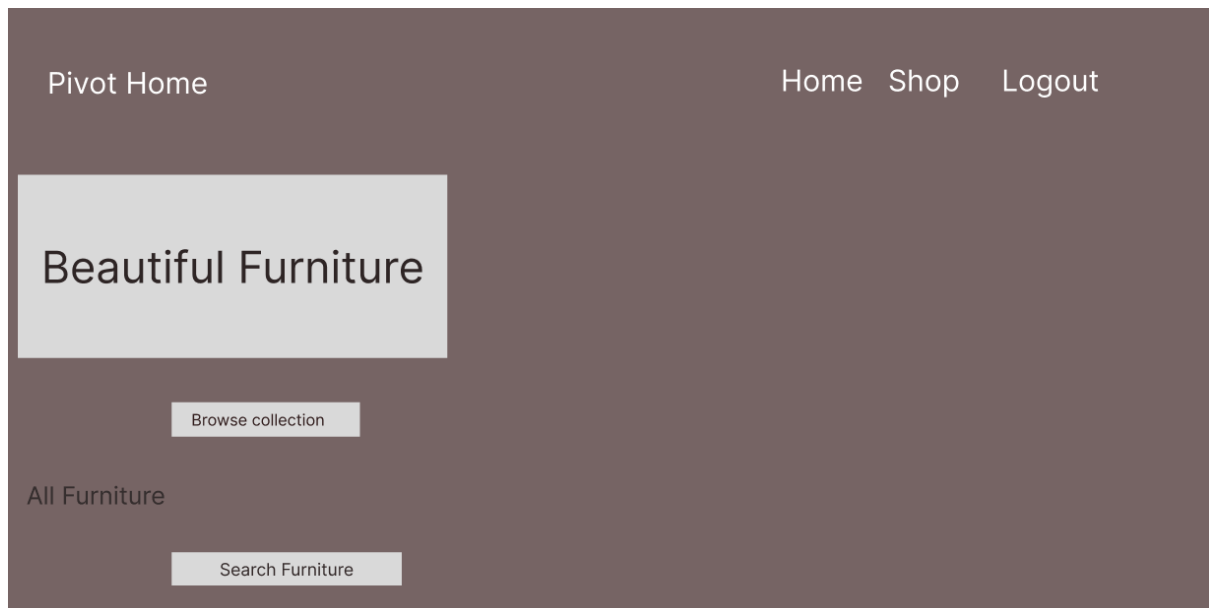
Fig 6: Successful Database Connection.

### **MY REFLECTION (Week1)**

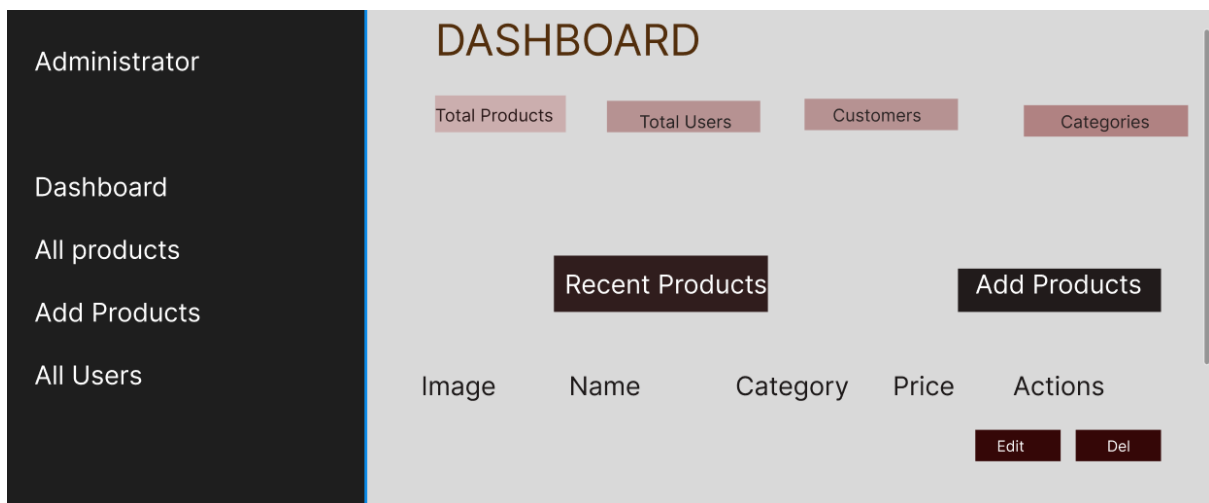
I learnt how to connect a system to a database .I used technologies such as PHP ,MySQL , XAMPP, HTML,CSS and Java script . Some of the issues I encountered are When one logs in it was taking you to a blank white page but I later found out I was being redirected to a page that didn't exist.

The screenshots demonstrate: Fig 1 -Successful XAMPP Installation. Fig 2 - Apache and MySQL running .Fig 3 - XAMPP Dashboard running on <http://localhost/>.Fig 4&5 -Test page file inside XAMPP htdocs folder and first successful PHP page. Fig 6 - Successful database connection.

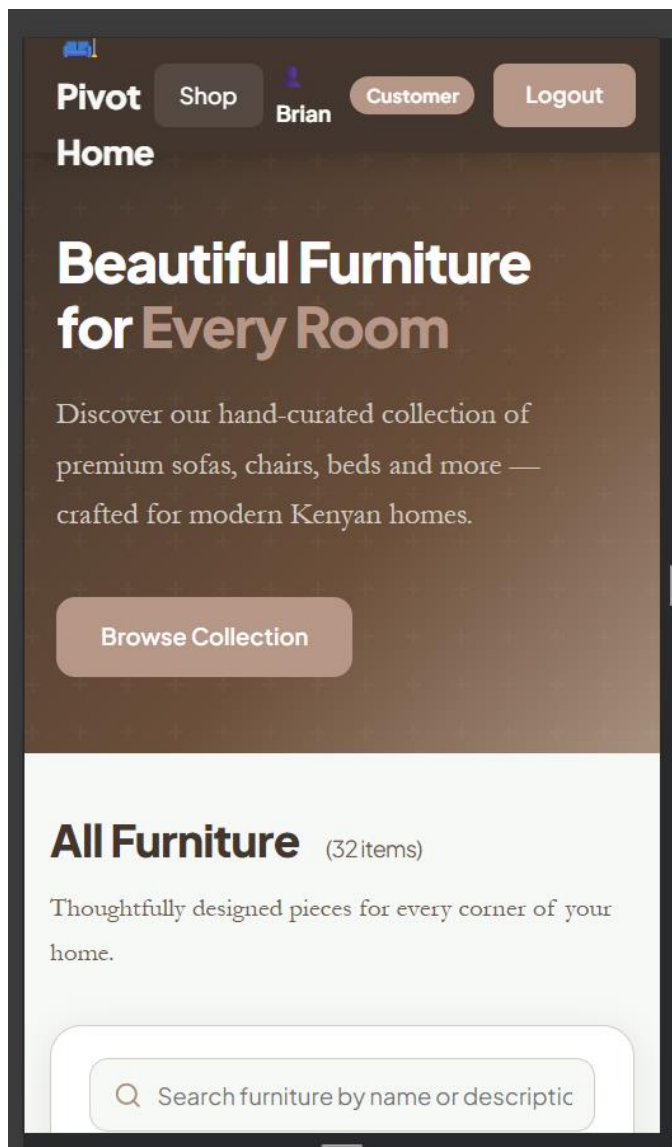
## Week 2: Wireframes and GUI Design



**Fig 1:** Wireframe showcasing nav bar shop now



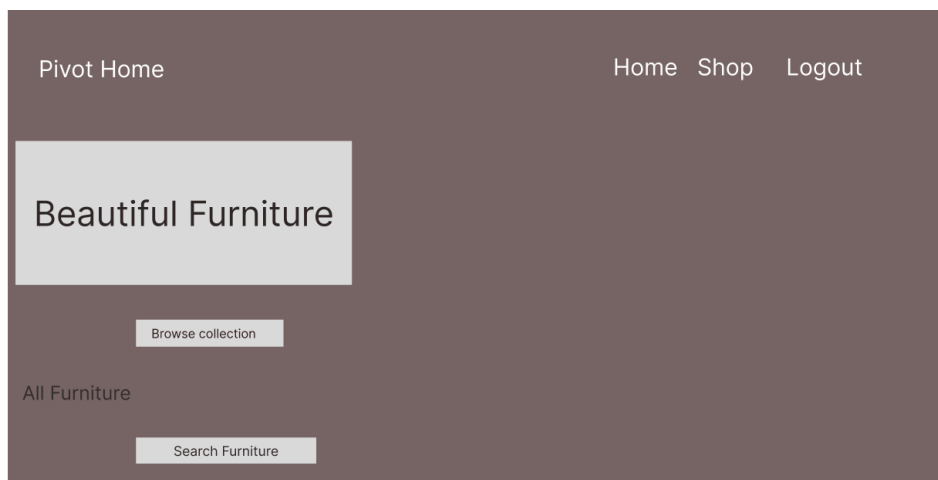
**Fig 2:** Dashboard layout design



**Fig 3: Mobile View Mockup**



**Fig 4: Color Theme**



**Fig 5: GUI Prototype in Figma**

### My reflection (Week 2)

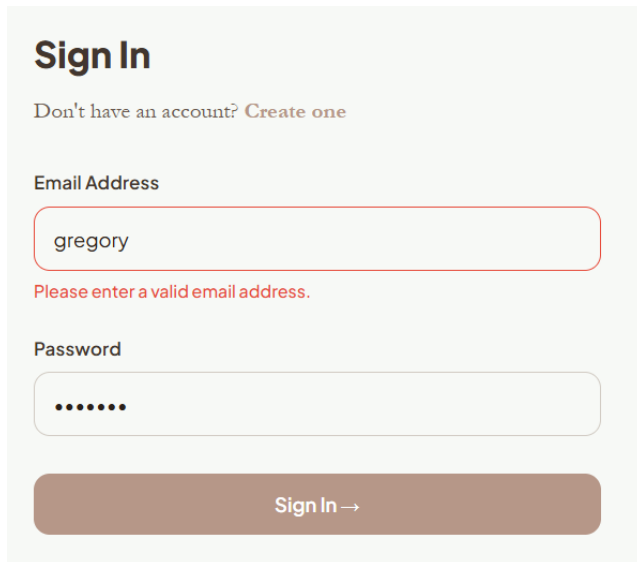
I chose the design since it was really beautiful and it shows the nature of raw materials used in the designing of furniture. I considered user experience by making it simple to navigate through and adding a search bar to find items easily and also put them into categories.

The navigation flow is such that customers can login and later logout while the admin has his own dashboard. The intended system users are the admin, guests and customers



# Week 3: JavaScript and PHP Basics

## Frontend Interaction and Backend Foundations



**Sign In**

Don't have an account? [Create one](#)

Email Address

gregory

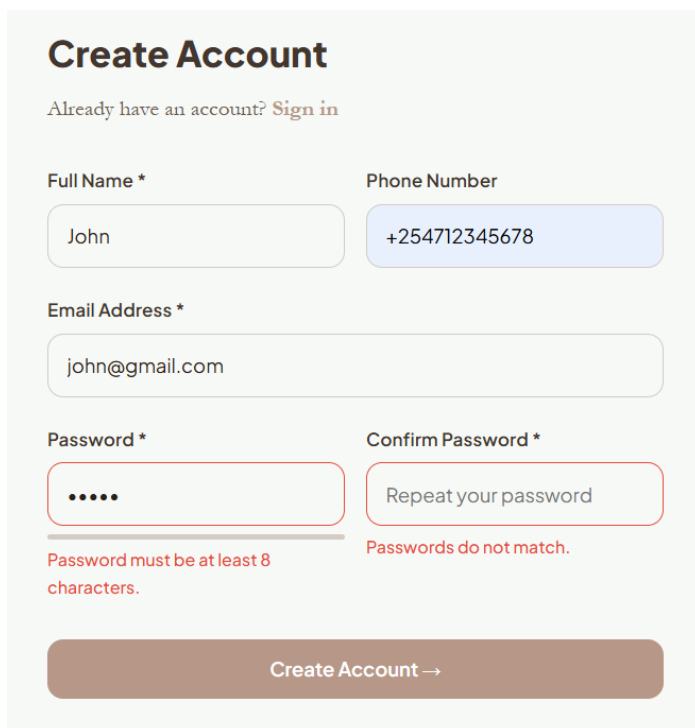
Please enter a valid email address.

Password

.....

Sign In →

**Fig 1: JavaScript Form Validation**



**Create Account**

Already have an account? [Sign in](#)

Full Name \*

John

Phone Number

+254712345678

Email Address \*

john@gmail.com

Password \*

.....

Confirm Password \*

Repeat your password

Password must be at least 8 characters.

Passwords do not match.

Create Account →

**Fig 2: Password Strength Checker**

```

admin > index.php
1  <?php
2  require_once __DIR__ . '/includes/auth.php';
3  if (isLoggedIn()) {
4      if (isAdmin()) {
5          header('Location: /furniture-store/admin/dashboard.php');
6      } else {
7          header('Location: /furniture-store/index.php');
8      }
9  } else {
10     header('Location: /furniture-store/login.php');
11 }
12 exit;
13

```

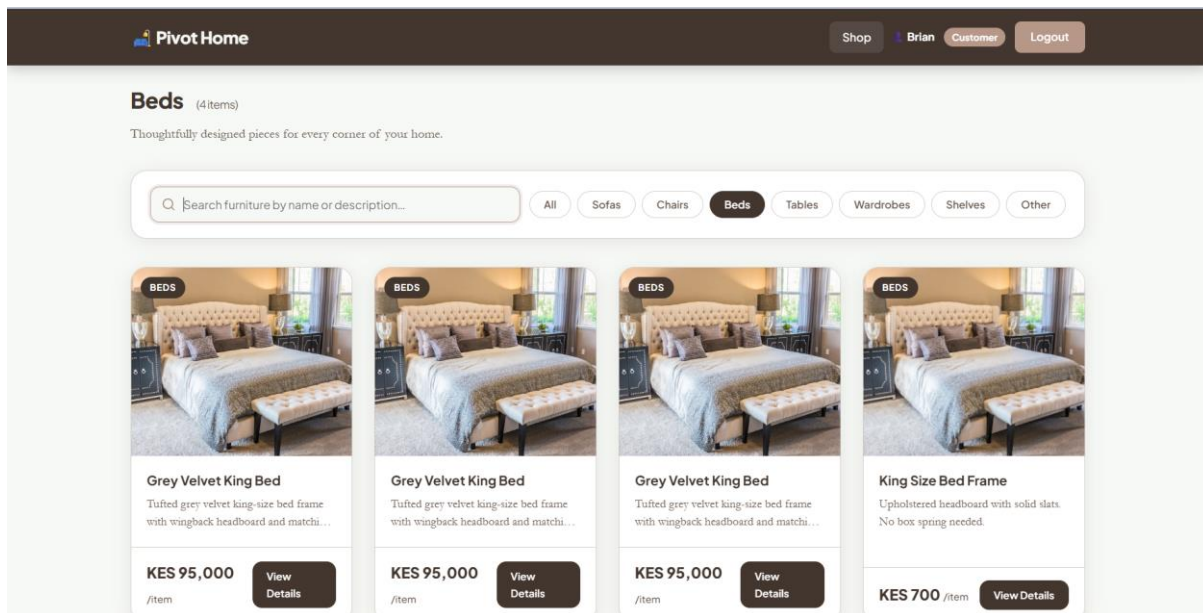
**Fig 3: PHP Syntax Practice**

```

includes > db.php > ...
1  <?php
2  define('DB_HOST', 'localhost');
3  define('DB_USER', 'root');
4  define('DB_PASS', '');
5  define('DB_NAME', 'furniture_store');
6
7  $conn = new mysqli(DB_HOST, DB_USER, DB_PASS, DB_NAME);
8
9  if ($conn->connect_error) {
10     die('<div style="font-family:sans-serif;padding:40px;background:#fff3cd;border:1px solid #ffc107;">
11         <h2>⚠ Database Connection Failed</h2>
12         <p>Please ensure MySQL is running and run <a href="/furniture-store/setup/install.php">setup/install
13         <p><code>' . $conn->connect_error . '</code></p>
14     </div>');
15 }
16
17 $conn->set_charset("utf8mb4");
18

```

**Fig 4: Database Connection Script**



**Fig 5: Dynamic User Input Handling**

### **My Reflection(Week 3)**

Frontend and backend interact by JavaScript running in the browser sends requests to PHP files on the server. PHP processes those requests, queries MySQL, and returns data which JavaScript then uses to update the page. **Database connection matters** because without the mysqli connection in db.php no page in Pivot Home can display products, validate login credentials, or process orders.

The purpose of validation is to ensure one does not submit invalid data. Technologies used are the HTML and CSS for the structure of the interface while PHP processes user requests server-side.

# WEEK 4

## Server-Side Components and Backend Development Foundations

**Sign In**

Don't have an account? [Create one](#)

Email Address

Password

Sign In →

Fig 4.1 Login form

**Sign In**

Don't have an account? [Create one](#)

Email Address

Please enter a valid email address.

Password

Password is required.

Sign In →

Fig 4.2 Form Validation

|                                                                                            |                  |             |
|--------------------------------------------------------------------------------------------|------------------|-------------|
|  xampp  | 02/10/2025 19:45 | File folder |
|  htdocs | 29/05/2026 09:55 | File folder |

| furniture-store |                  | 22/05/2026 09:25 | File folder         |      |
|-----------------|------------------|------------------|---------------------|------|
| 📁               | (includes,assets | 22/05/2026 09:25 | File folder         |      |
| 📁               | admin            | 22/05/2026 09:25 | File folder         |      |
| 📁               | assets           | 22/05/2026 09:25 | File folder         |      |
| 📁               | includes         | 22/05/2026 09:25 | File folder         |      |
| 📁               | setup            | 22/05/2026 09:25 | File folder         |      |
| 📄               | .htaccess        | 22/05/2026 09:25 | HTACCESS File       | 1 KB |
| 📄               | index            | 22/05/2026 09:25 | PHP Source File     | 6 KB |
| 📄               | login            | 22/05/2026 09:25 | PHP Source File     | 4 KB |
| 📄               | logout           | 22/05/2026 09:25 | PHP Source File     | 1 KB |
| 📄               | product          | 22/05/2026 09:25 | PHP Source File     | 5 KB |
| 📄               | README           | 22/05/2026 09:25 | Markdown Source ... | 6 KB |
| 📄               | register         | 22/05/2026 09:25 | PHP Source File     | 5 KB |

Fig 4.3 Folder structure

```

1 <?php
2 define('DB_HOST', 'localhost');
3 define('DB_USER', 'root');
4 define('DB_PASS', '');
5 define('DB_NAME', 'furniture_store');
6
7 $conn = new mysqli(DB_HOST, DB_USER, DB_PASS, DB_NAME);
8
9 if ($conn->connect_error) {
10     die('<div style="font-family:sans-serif;padding:40px;background:#fff3cd;border:1px solid #ffc107;back
11         <h2>⚠ Database Connection Failed</h2>
12         <p>Please ensure MySQL is running and run <a href="/furniture-store/setup/install.php">setup/inst
13         <p><code>' . $conn->connect_error . '</code></p>
14         </div>');
15 }
16
17 $conn->set_charset("utf8mb4");
18

```

Fig 4.4 Database connection

```

1  <?php
2  $adminTitle = 'Add New Product';
3  require_once __DIR__ . '/admin_header.php';
4
5  $categories = ['Sofas','Chairs','Beds','Tables','Wardrobes','Shelves','Other'];
6  $error = '';
7
8  if ($_SERVER['REQUEST_METHOD'] === 'POST') {
9      $name      = trim($_POST['name'] ?? '');
10     $desc       = trim($_POST['description'] ?? '');
11     $price      = $_POST['price'] ?? '';
12     $cat        = $_POST['category'] ?? '';
13     $imageUrl   = trim($_POST['image_url'] ?? '');
14
15     // Handle file upload
16     if (isset($_FILES['image_file']) && $_FILES['image_file']['error'] === UPLOAD_ERR_OK) {
17         $ext = strtolower(pathinfo($_FILES['image_file']['name'], PATHINFO_EXTENSION));
18         $allowed = ['jpg','jpeg','png','webp','gif'];
19         if ( (global variable) string $filename
20             $filename = 'product_' . time() . '_' . rand(1000,9999) . '.' . $ext;
21             $dest = __DIR__ . '/../assets/uploads/' . $filename;
22             if (move_uploaded_file($_FILES['image_file']['tmp_name'], $dest)) {
23                 $imageUrl = '../furniture-store/assets/uploads/' . $filename;
24             }
25         }
26     }
27
28     // Validate
29     if (strlen($name) < 3) $error = 'Product name must be at least 3 characters.';
30     elseif (!is_numeric($price) || $price <= 0) $error = 'Enter a valid price.';
31     elseif (!in_array($cat, $categories)) $error = 'Please select a valid category.';
32     elseif (strlen($desc) < 10) $error = 'Description must be at least 10 characters.';
33     else {
34         $stmt = $conn->prepare("INSERT INTO products (name, description, price, image_url, cat
35         $stmt->bind_param('ssdss', $name, $desc, $price, $imageUrl, $cat);

```

Fig 4.5 Code Snippets

# WEEK 5

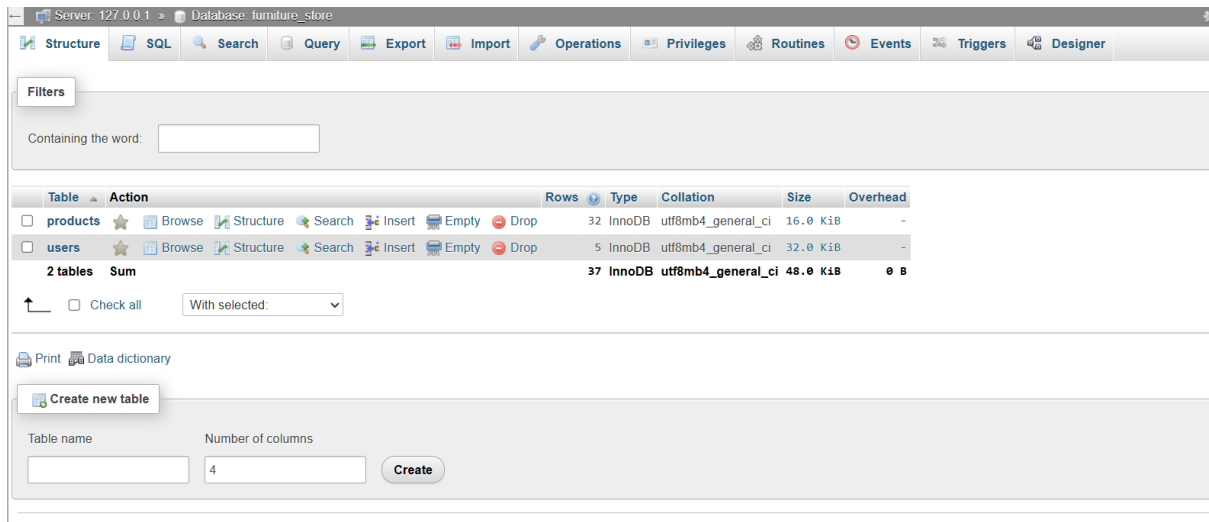


Fig 5.1 Database screenshot

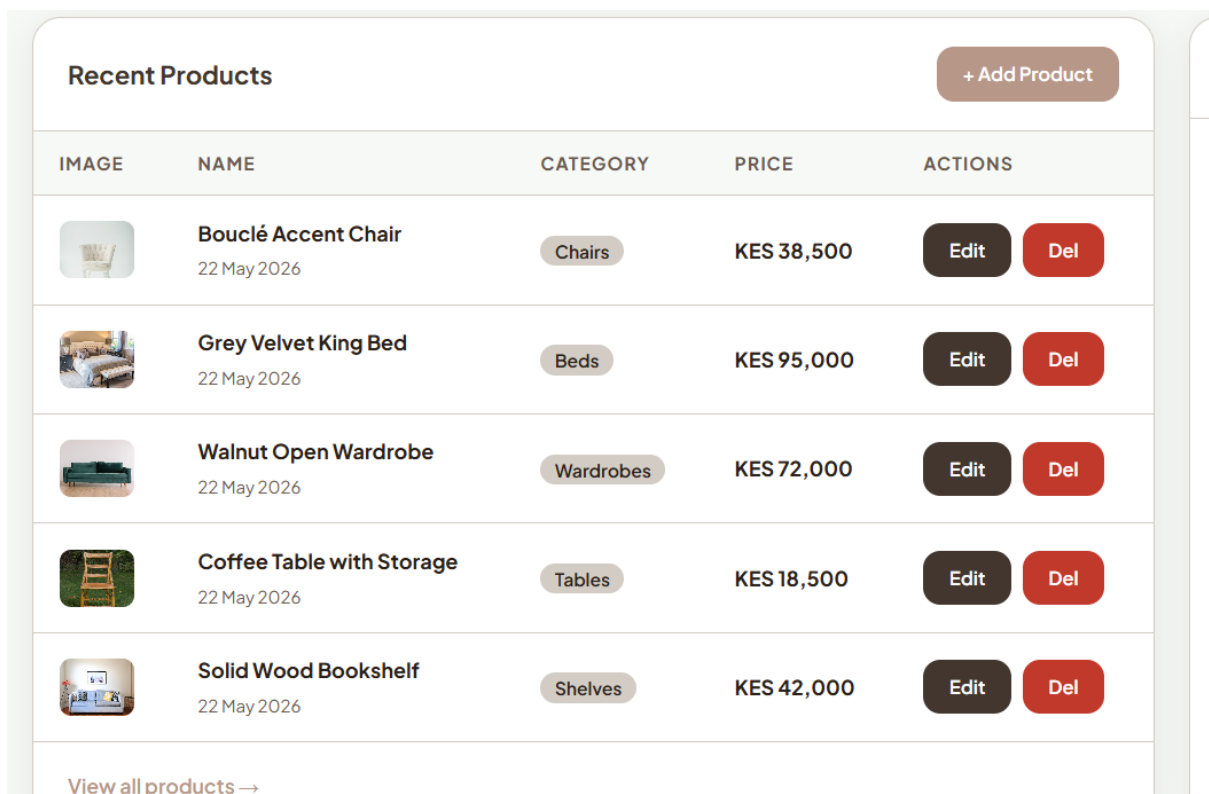


Fig 5.2 Crud Operations

```

1  <?php
2  require_once __DIR__ . '/includes/auth.php';
3  require_once __DIR__ . '/includes/db.php';
4  redirectIfLoggedIn();
5
6  $error = $success = '';
7
8  if ($_SERVER['REQUEST_METHOD'] === 'POST') {
9      $name = trim($_POST['name'] ?? '');
10     $email = trim($_POST['email'] ?? '');
11     $phone = trim($_POST['phone'] ?? '');
12     $password = $_POST['password'] ?? '';
13     $confirm = $_POST['confirm_password'] ?? '';
14
15     // Server-side validation
16     if (strlen($name) < 2) {
17         $error = 'Full name must be at least 2 characters.';
18     } elseif (!filter_var($email, FILTER_VALIDATE_EMAIL)) {
19         $error = 'Please enter a valid email address.';
20     } elseif (strlen($password) < 8 || !preg_match('/[A-Z]/', $password) || !preg_match('/[0-9]/', $password)) {
21         $error = 'Password must be 8+ characters with at least one uppercase letter and one number.';
22     } elseif ($password !== $confirm) {
23         $error = 'Passwords do not match.';
24     } else {
25         // Check email exists
26         $stmt = $conn->prepare("SELECT id FROM users WHERE email = ?");
27         $stmt->bind_param('s', $email);
28         $stmt->execute();
29         $stmt->store_result();
30     }
31 }

```

Fig 5.3 Login database intergration

Browse

Structure

SQL

Search

Insert

Export

Import

Privileges

Operations

Triggers

Showing rows 0 - 24 (32 total, Query took 0.0003 seconds.)

SELECT \* FROM `products`

Profiling

Edit inline

Edit

Explain SQL

Create PHP code

Refresh

1

>

>>

Show all

Number of rows: 25

Filter rows: Search this table

Sort by key: None

Extra options

|                          |                                                            | id | name                    | description                                           | price  | image_url                                             | category | created_at          |
|--------------------------|------------------------------------------------------------|----|-------------------------|-------------------------------------------------------|--------|-------------------------------------------------------|----------|---------------------|
| <input type="checkbox"/> | <div><div>Edit</div><div>Copy</div><div>Delete</div></div> | 1  | Modern Velvet Sofa      | Luxurious velvet upholstery with solid wood frame.... | 599.99 | https://images.unsplash.com/photo-1555041469-a586c... | sofas    | 2026-05-22 08:47:24 |
| <input type="checkbox"/> | <div><div>Edit</div><div>Copy</div><div>Delete</div></div> | 2  | Rustic Oak Dining Table | Handcrafted solid oak table, seats 6-8 people. Dur... | 899.99 | https://images.unsplash.com/photo-1577140917170-28... | tables   | 2026-05-22 08:47:24 |
| <input type="checkbox"/> | <div><div>Edit</div><div>Copy</div><div>Delete</div></div> | 3  | Ergonomic Office Chair  | Adjustable height with lumbar support. Mesh back f... | 249.99 | https://images.unsplash.com/photo-1580480055273-22... | chairs   | 2026-05-22 08:47:24 |
| <input type="checkbox"/> | <div><div>Edit</div><div>Copy</div><div>Delete</div></div> | 4  | Minimalist Coffee Table | Clean lines with walnut finish. Includes hidden st... | 299.99 | https://images.unsplash.com/photo-1533090481720-85... | tables   | 2026-05-22 08:47:24 |
| <input type="checkbox"/> | <div><div>Edit</div><div>Copy</div><div>Delete</div></div> | 5  | Scandinavian Armchair   | Wool blend fabric with oak legs. Light and comfort... | 349.99 | https://images.unsplash.com/photo-1567538096630-e0... | chairs   | 2026-05-22 08:47:24 |
| <input type="checkbox"/> | <div><div>Edit</div><div>Copy</div><div>Delete</div></div> | 6  | King Size Bed Frame     | Upholstered headboard with solid slats. No box spr... | 699.99 | https://images.unsplash.com/photo-1505693416388-ac... | beds     | 2026-05-22 08:47:24 |

Fig 5.4 SQL complete Query



